National University of Computer and Emerging Sciences
Lahore Campus

Signals and Systems (EE2008)

Sessional-I Exam

Date: February 25, 2026

Course Instructor(s)

1. Dr. S. M. Sajid
2. Khalid Ijaz
3. Mohsin Yousuf (CM)

Total Time (Hrs):	1
Total Marks:	30
Total Questions:	3

Roll No Section Student Signature

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1. **Attempt all questions and remember to solve parts of the same question together.**
2. **Show all the steps with the help of labelled diagrams and mathematical equations.**

CLO # 01: Apply *fundamental concepts of signals and systems to classify and represent various types of signals and systems.*

Q1: An input-output system is defined by a differential equation. The output for an input $x_1(t)$ is $y_1(t)$ and similarly for an input $x_2(t)$, the output is $y_2(t)$. **[10]**

$$x_1 \to y_1 \quad \text{and} \quad x_2 \to y_2$$

If the system is linear i.e. it obeys *superposition (homogeneity and additivity)*, **show** that:

$$(D^N + a_1 D^{N-1} + \cdots + a_{N-1}D + a_N)y(t)$$
$$= (b_{N-M}D^M + b_{N-M+1}D^{M-1} + \cdots + b_{N-1}D + b_N)x(t)$$

Also, **interpret** values of M and N for practical systems.

CLO # 02: Analyze *continuous and discrete-time Linear Time-Invariant (LTI) systems using convolution operations.*

Q2: The unit impulse response of an LTIC system is **[10]**

$$h(t) = u(t-3)$$

For this system, **find** and sketch its (zero-state) response $y(t)$ using *direct integration*, if the input is

$$x(t) = e^{-at}u(t), \qquad a > 0$$

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*CLO # 02: **Analyze** continuous and discrete-time Linear Time-Invariant (LTI) systems using convolution operations.*

Q3: **Find** and sketch the response $y(t) = x_1(t) * x_2(t)$ using graphical convolution for the following inputs. **[10]**